

# **Dell PowerProtect Data Domain Diagnostic: CVE-2026-26951**

Root Escalation via Stack-Based Buffer Overflow

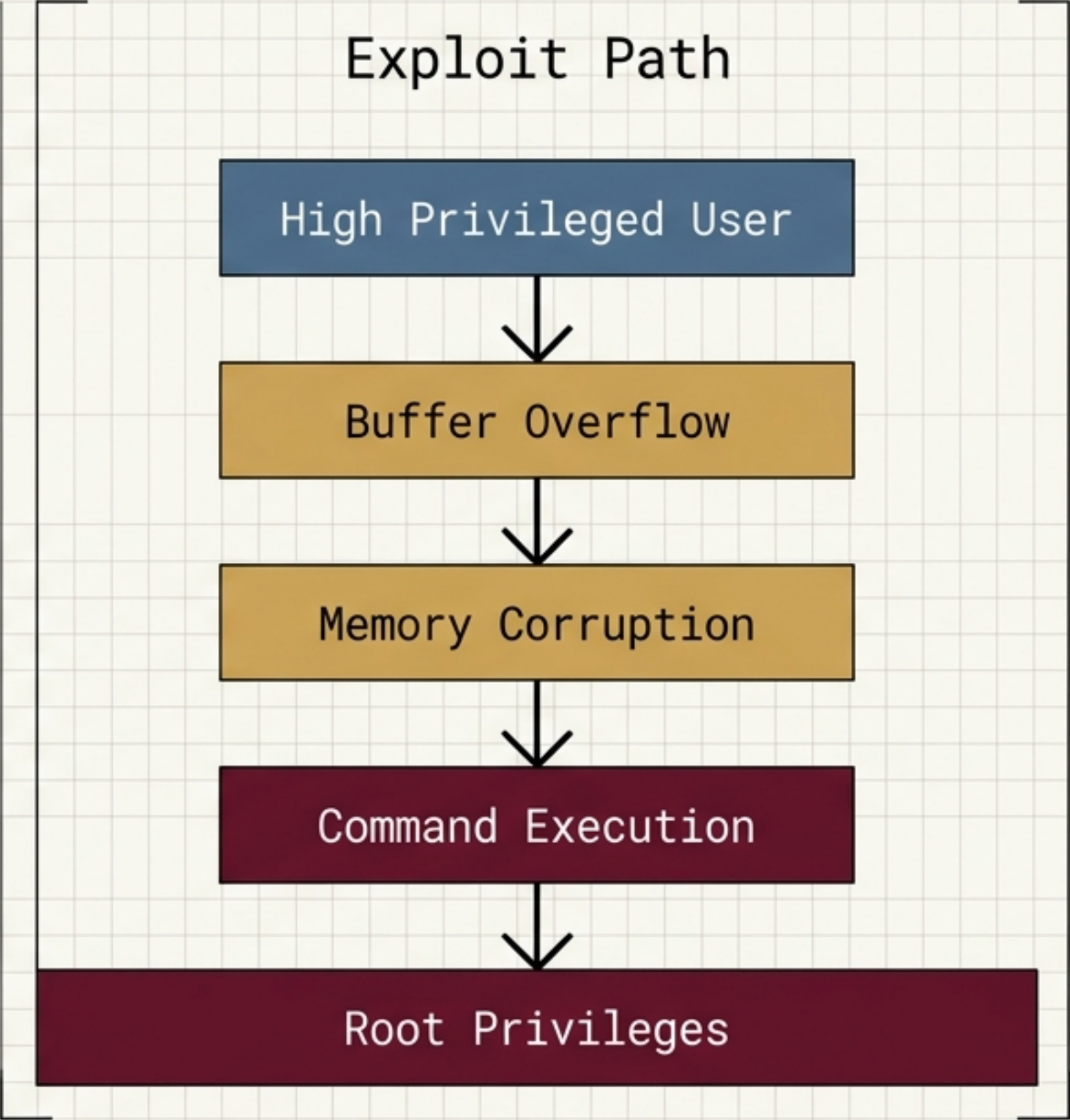
[CWE-121]

[CVSS v3.1: 6.7]

[Severity: MEDIUM]

# Threat Profile Matrix

|                                                                                     |                           |
|-------------------------------------------------------------------------------------|---------------------------|
| CVSS:3.1/AV:L/AC:L/PR:H/UI:N/S:U/C:H/I:H/A:H                                        |                           |
|    | Attack Vector: Local      |
|    | Attack Complexity: Low    |
|  | Privileges Required: High |
|  | User Interaction: None    |



Takeaway: A high-privileged local attacker can exploit this memory management flaw to execute arbitrary commands with absolute root control.

# The Escalation Choke Point

## Prerequisites

- ✓ Local Access Required
- ✓ High Privileges Required
- ✓ Access to Vulnerable Component
- ✗ No Network Access
- ✗ No User Interaction

## Impact



**Arbitrary  
Command  
Execution**



**Full Root  
Privileges**

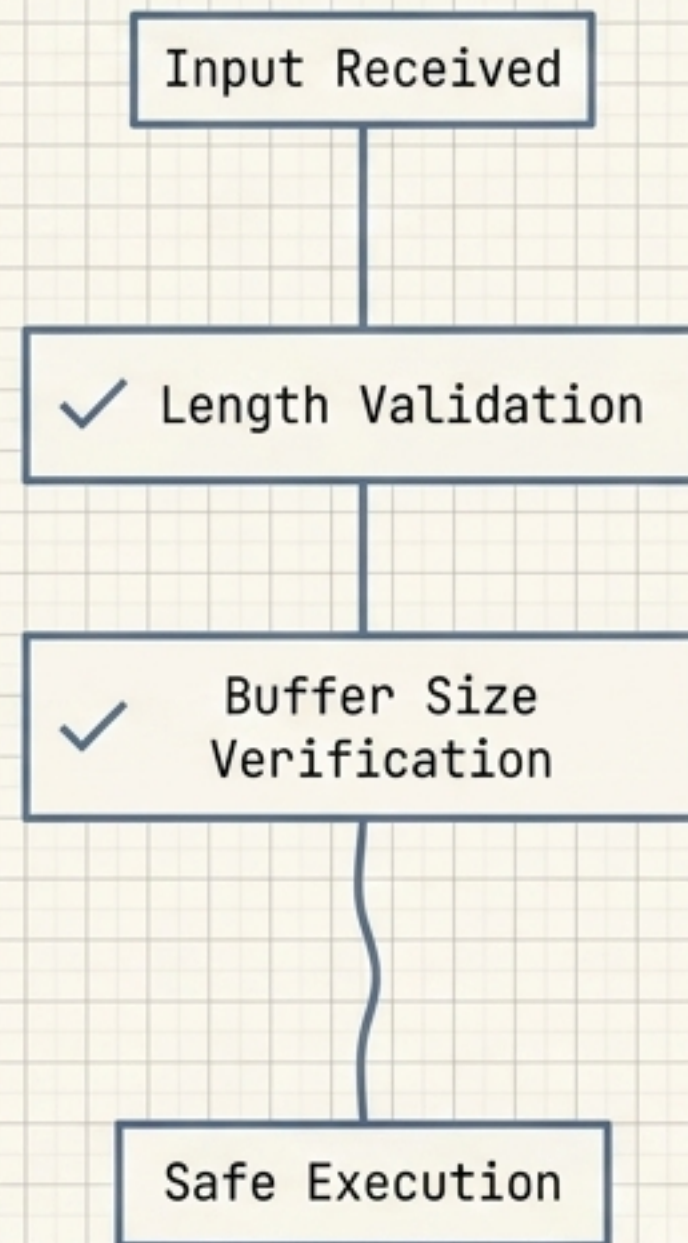
## Synthesis Insight

The Secondary Weapon: This is not an initial breach vector. A threat actor must already be inside the network and holding high privileges to use this. It is a fatal escalation choke point, not an open door.

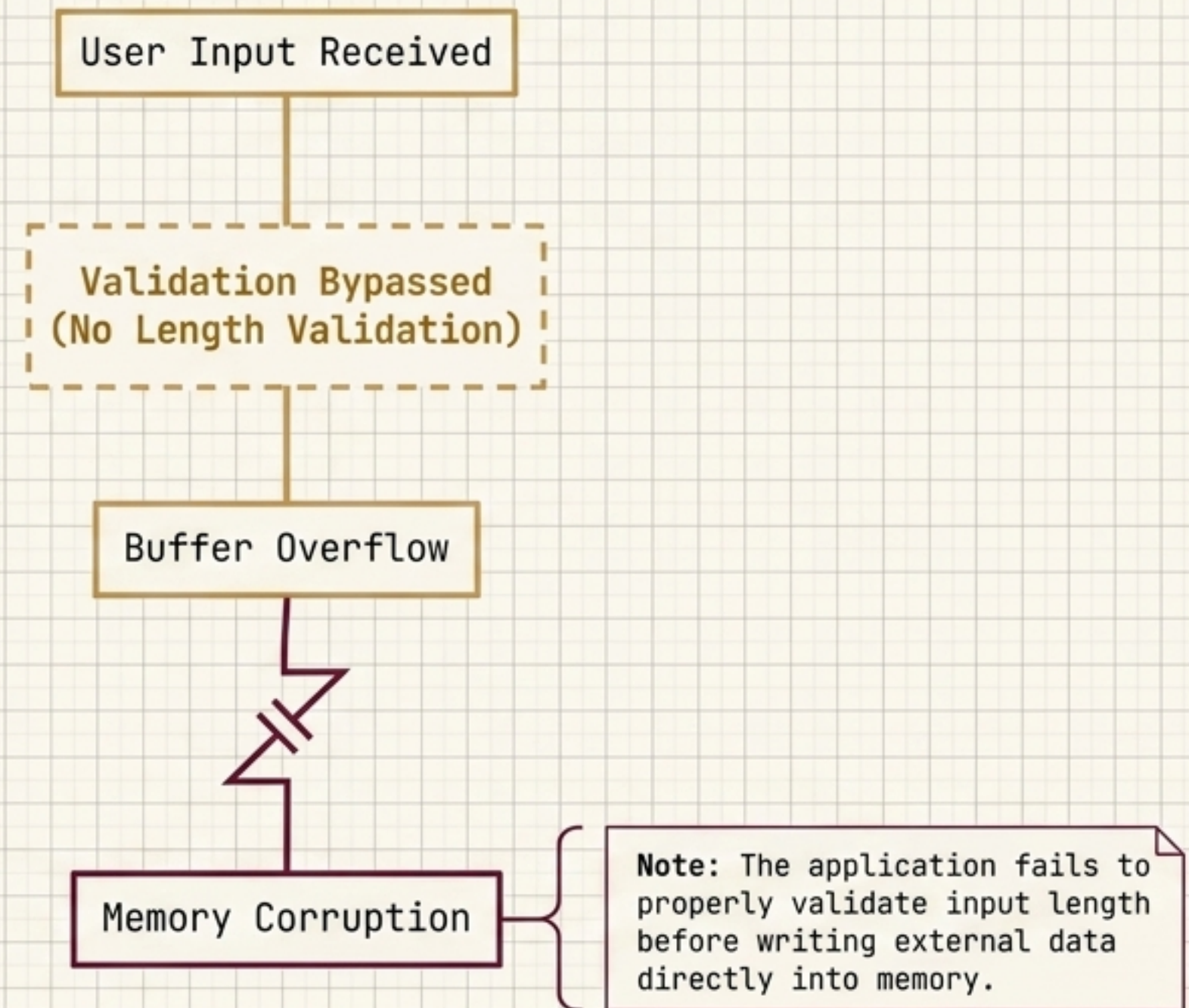
# Root Cause Analysis: CWE-121

## Secure vs. Vulnerable Behavior Matrix

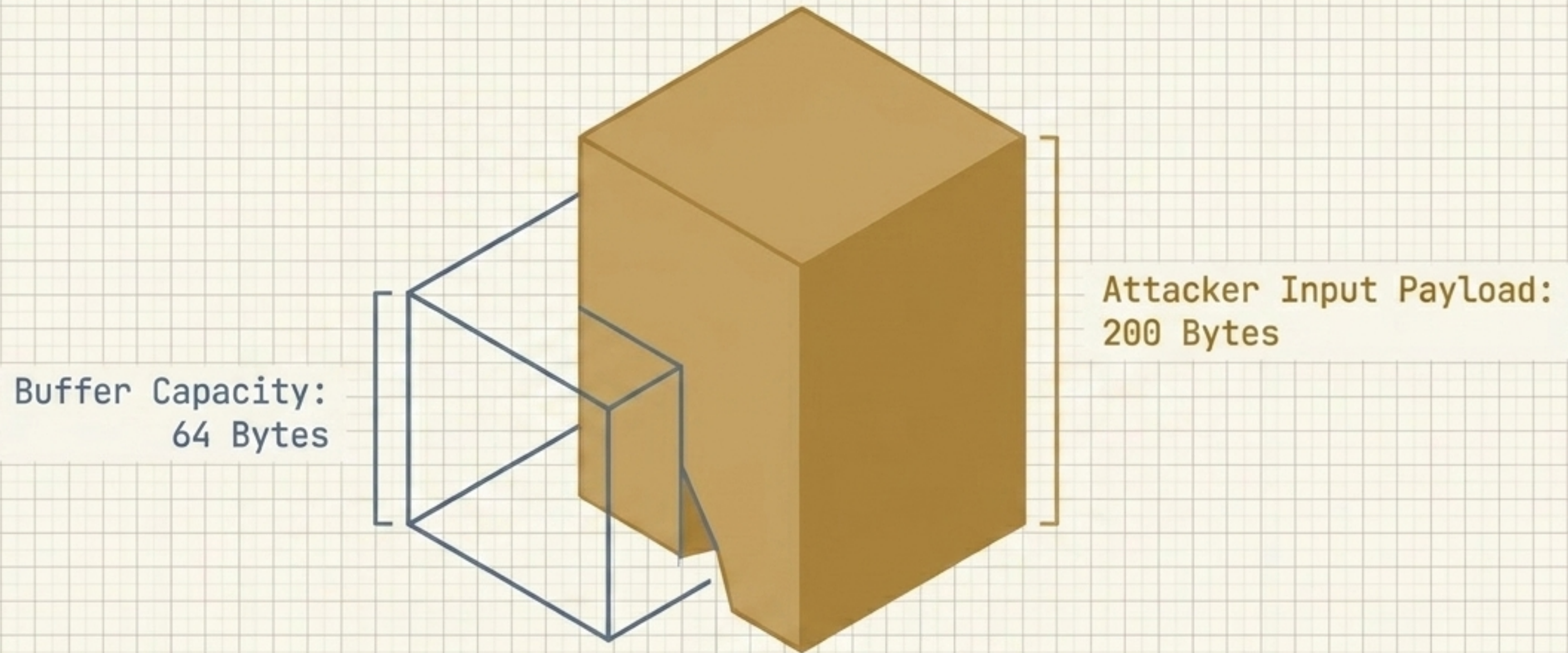
### Secure Behavior



### Vulnerable Behavior



# The Mechanics of a Stack Buffer Overflow



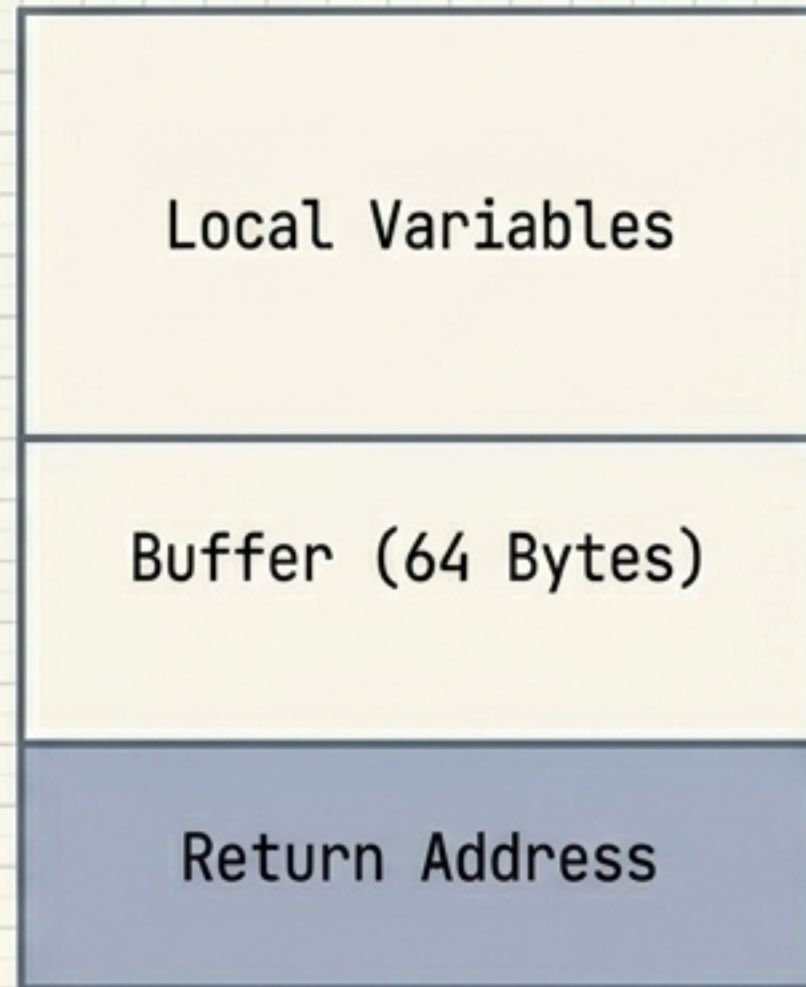
1. Volume Exceeds  
Capacity

2. Adjacent Memory is  
Overwritten by Excess Data

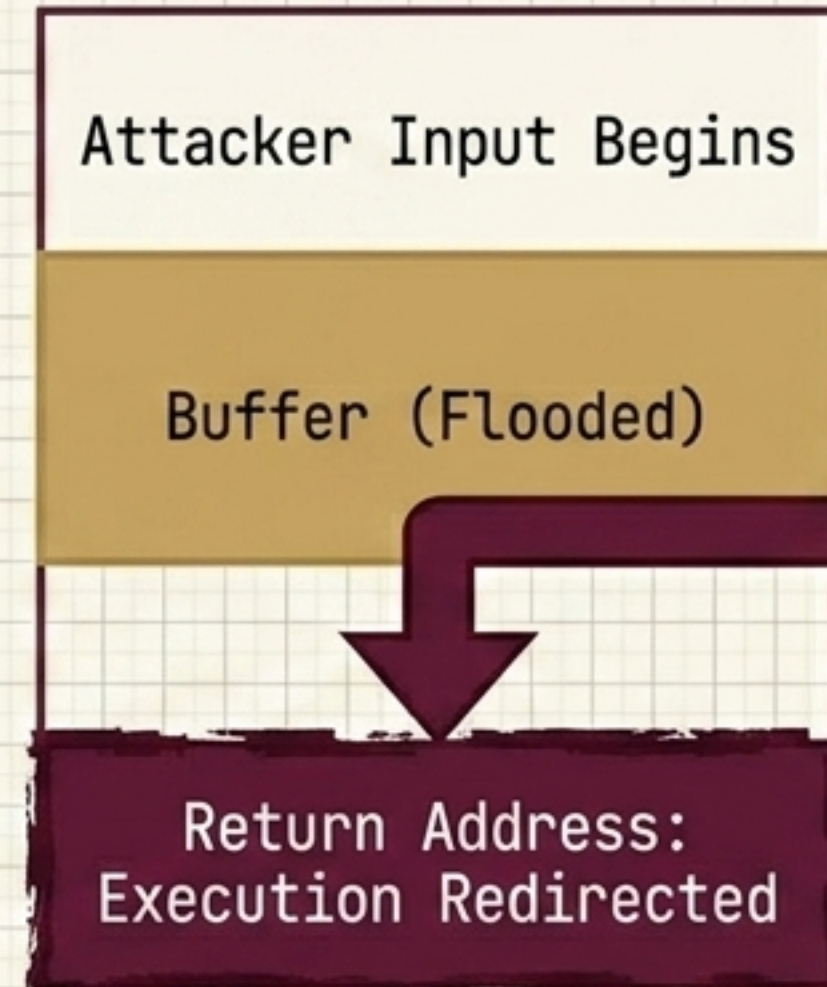
3. Program Control is  
Forcibly Altered

# Memory Layout Structural Comparison

Normal Execution

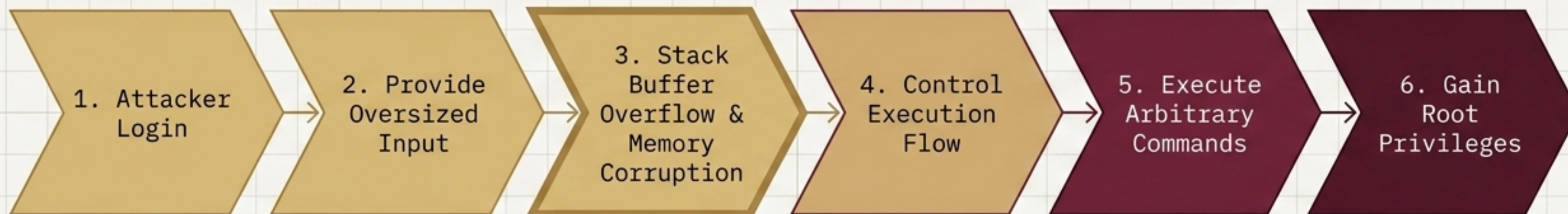


Overflow Condition

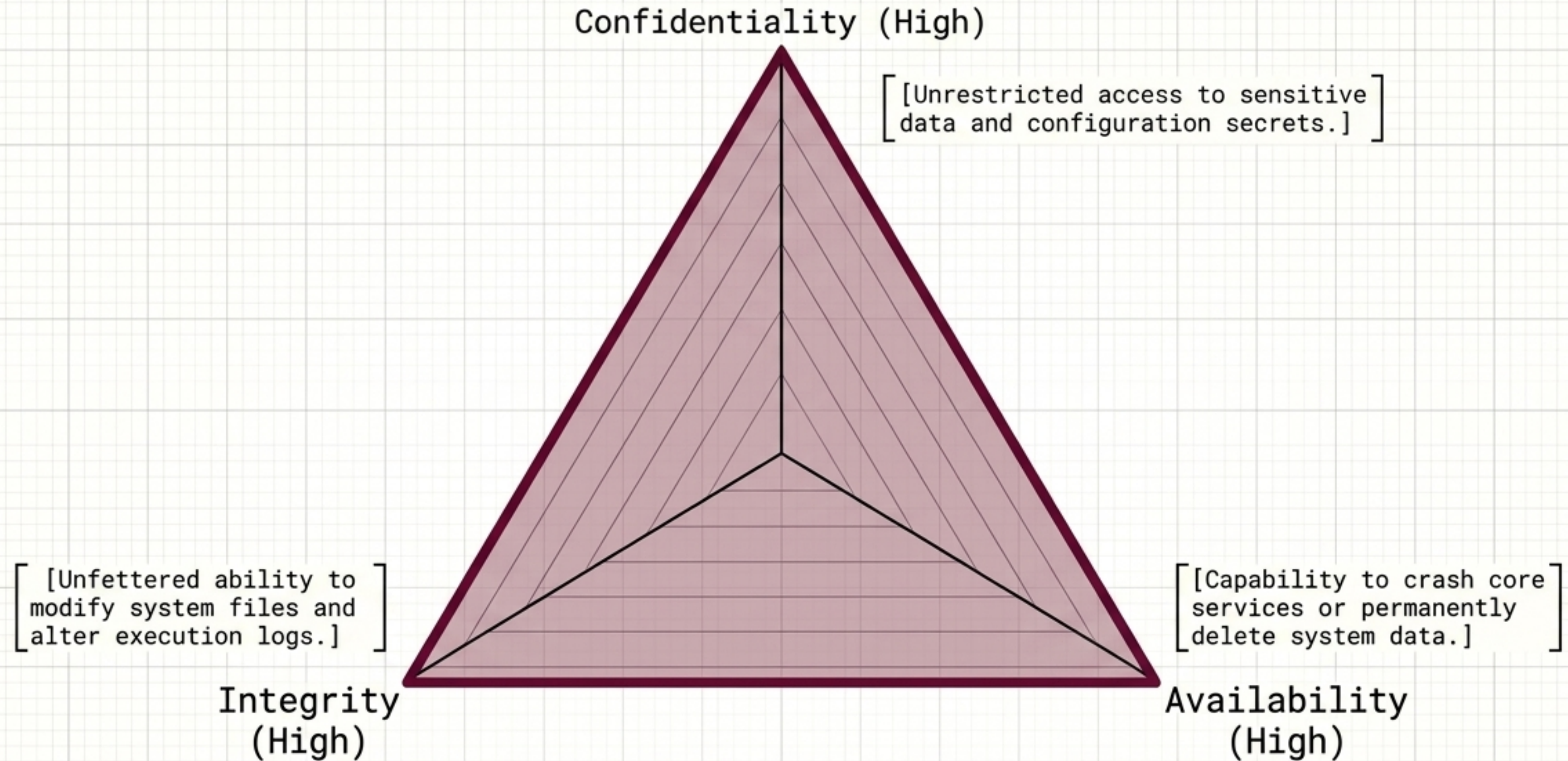


Takeaway: By flooding the local variables, the attacker forces their own executable path over the system's intended return address.

# Exploitation Flow Timeline

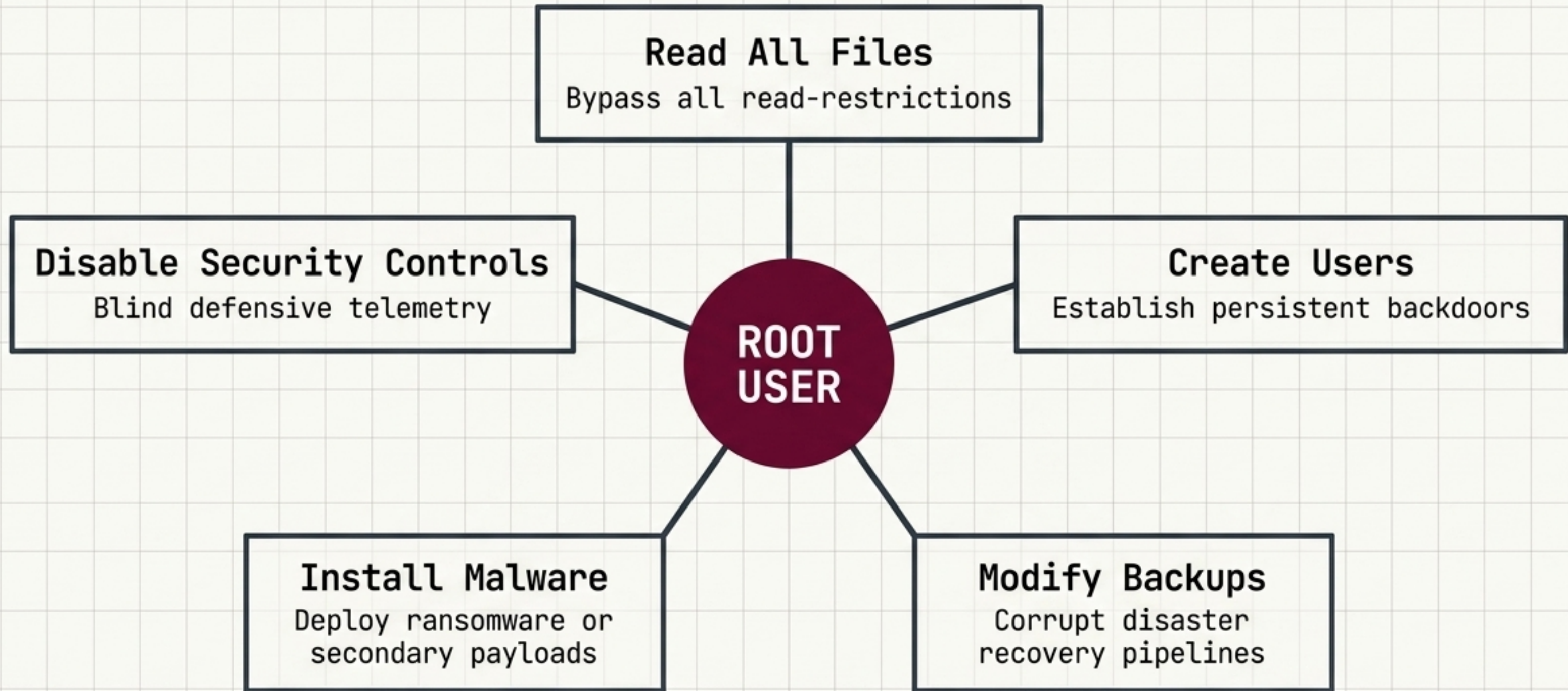


# The Technical Impact Profile



This total compromise is the mathematical driver behind the heavy impact weighting in the CVSS 6.7 score.

# Root Access Blast Radius



Root access effectively nullifies all system boundaries, granting absolute operational control over the Dell Data Domain environment.

# MITRE ATT&CK

## Post-Exploitation Mapping

### Privilege Escalation

[T1068]

Exploitation for Privilege Escalation. The primary mechanism of the attack.

### Defense Evasion / Privilege Escalation

[T1548]

Abuse Elevation Control Mechanism. Bypassing standard user constraints to achieve root.

### Defense Evasion

[T1055]

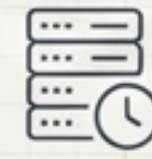
Process Injection. Anticipated follow-on behavior to maintain stealth and persistence post-root.

# Defensive Telemetry & Detection Opportunities



## Real-Time Endpoint Detection (EDR)

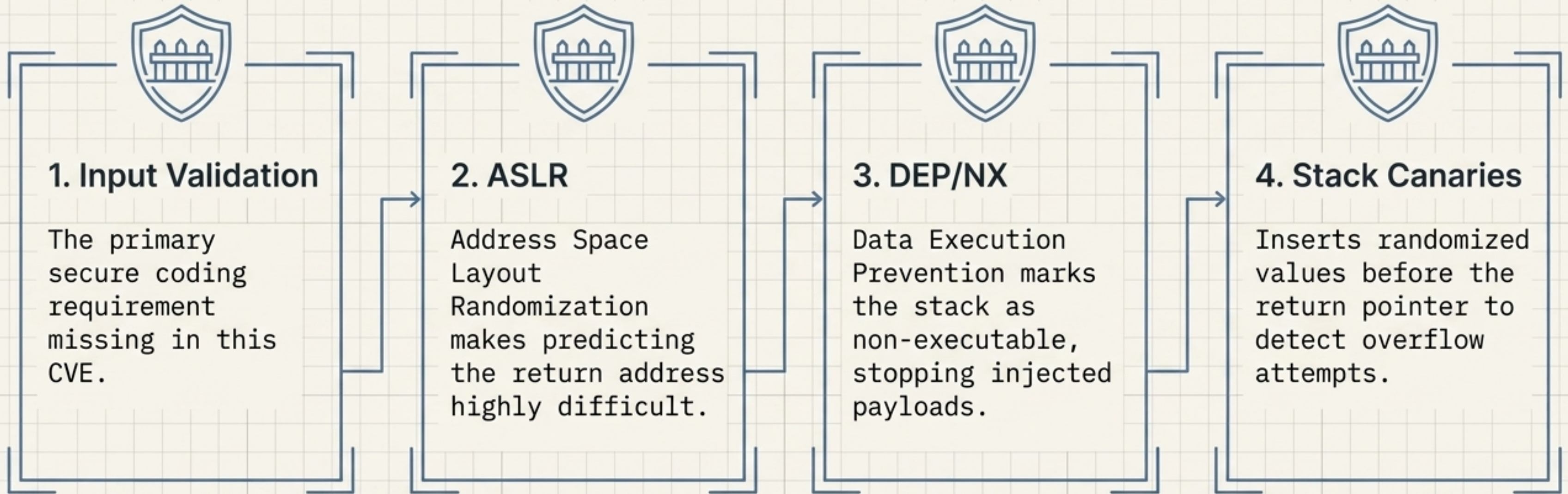
- Monitor for **Privilege Escalation** anomalies originating from local execution.
- Flag anomalous **Memory Corruption** patterns.
- Alert on **Suspicious Process Behavior** (unexpected child processes).



## Historical System Logs & Artifacts

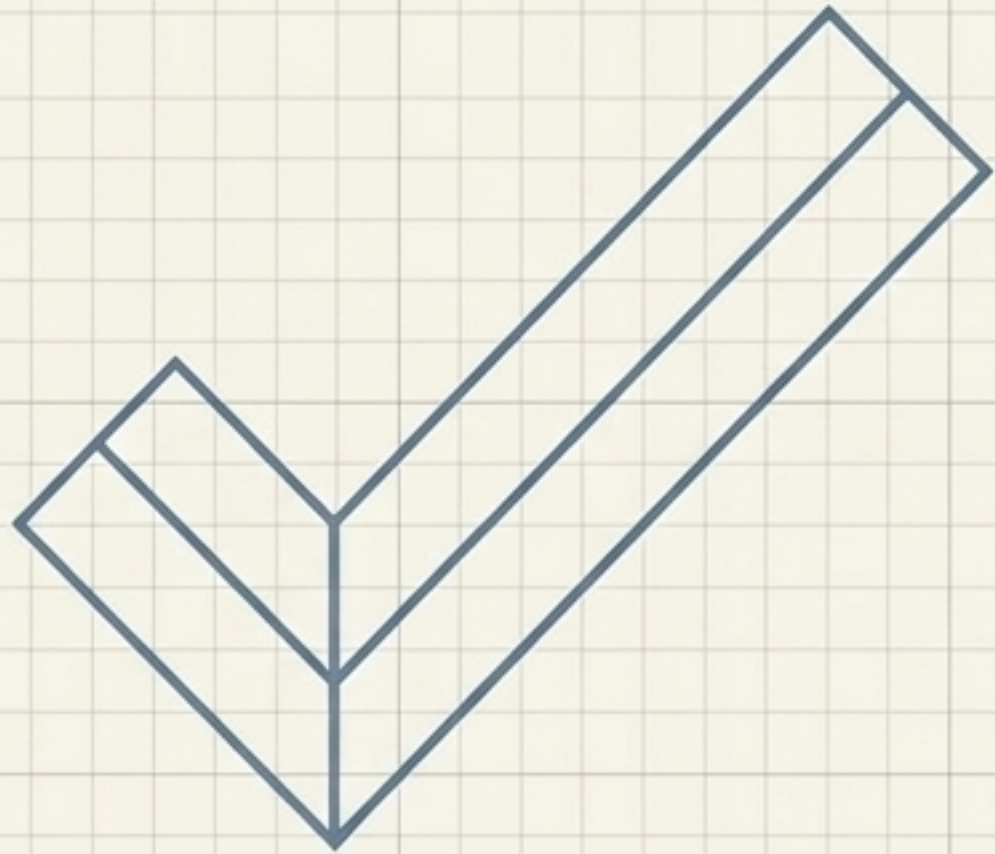
- Watch for **System Segmentation Faults** and **Application Crashes**.
- Analyze generated **Core Dumps** for overflow patterns.
- Flag any **Root Shell Activity** or Unexpected **Service Restarts**.

# Structural OS & Code-Level Defenses



# Eradication: Dell Verified Upgrade Paths

Mitigation requires patching PowerProtect Data Domain to one of the following validated releases:



Version 8.6.1.10+

Version 8.7.0.0+

Version 8.3.1.30+

Version 7.13.1.70+

**Action Directive:** Prioritize patching environments where highly privileged local access is broadly distributed.

## Research Conclusion & Final Readout

CVE-2026-26951 is a stack-based buffer overflow affecting Dell PowerProtect Data Domain. A **critical failure** in input validation allows local, high-privileged attackers to **corrupt memory** and hijack system execution flow. While the requirement for existing local privileges limits its use as an initial breach vector, successful exploitation transforms partial access into complete, unrestricted root system compromise. Immediate patching to validated firmware versions is required.